The worst thesis in the world

Thesis by
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In Partial Fulfillment of the Requirements for the
Degree of
[Name of Degree]

CALIFORNIA INSTITUTE OF TECHNOLOGY
Pasadena, California

2026
Defended May 15

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ACKNOWLEDGEMENTS

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ABSTRACT

[This abstract must provide a succinct and informative condensation of your work. Candidates are welcome to prepare a lengthier abstract for inclusion in the dissertation, and provide a shorter one in the CaltechTHESIS record.]

PUBLISHED CONTENT AND CONTRIBUTIONS

First author contributions: [20], [14]

Major contributions: [6], [15], [21]

SpECTRE core developer and maintainer of Cauchy-Characteristic Extraction (CCE) module: [11], [12], [22], [19], [3], [8], [9], [17], [4], [10], [7], [23], [16], [18], [2], [1], [13], [24], [25], [5]

[1] Yitian Chen et al. Improved frequency spectra of gravitational waves with memory in a binary-black-hole simulation. *Phys. Rev. D*, 110(6):064049, 2024.

[2] Teagan A. Clarke et al. Toward a self-consistent framework for measuring black hole ringdowns. *Phys. Rev. D*, 109(12):124030, 2024.

[3] Guido Da Re et al. Modeling the BMS transformation induced by a binary black hole merger. *Phys. Rev. D*, 111(12):124019, 2025.

[4] Marina De Amicis et al. Late-Time Tails in Nonlinear Evolutions of Merging Black Holes. *Phys. Rev. Lett.*, 135(17):171401, 2025.

[5] Nils Deppe et al. Simulating magnetized neutron stars with discontinuous Galerkin methods. *Phys. Rev. D*, 105(12):123031, 2022.

[6] Nils Deppe et al. Binary neutron star mergers using a discontinuous Galerkin-finite difference hybrid method. *Class. Quant. Grav.*, 41(24):245002, 2024.

[7] Nils Deppe, Lavinia Heisenberg, Henri Inchauspé, Lawrence E. Kidder, David Maibach, Sizheng Ma, Jordan Moxon, Kyle C. Nelli, William Throwe, and Nils L. Vu. Echoes from beyond: Detecting gravitational-wave quantum imprints with LISA. *Phys. Rev. D*, 111(12):124035, 2025.

[8] Nils Deppe, Lavinia Heisenberg, Lawrence E. Kidder, David Maibach, Sizheng Ma, Jordan Moxon, Kyle C. Nelli, William Throwe, and Nils L. Vu. Signatures of quantum gravity in gravitational wave memory. *Phys. Rev. D*, 112(2):024016, 2025.

[9] Leda Gao et al. Robustness of extracting quasinormal mode information from black hole merger simulations. *Phys. Rev. D*, 112(2):024025, 2025.

[10] Matthew Giesler et al. Overtones and nonlinearities in binary black hole ringdowns. *Phys. Rev. D*, 111(8):084041, 2025.

[11] Sarah Habib et al. Error quantification and comparison of binary neutron star gravitational waveforms from numerical relativity codes. 9 2025.

[12] Guillermo Lara et al. Signatures from metastable oppositely-charged black hole binaries in scalar Gauss-Bonnet gravity. 5 2025.

[13] Guillermo Lara, Harald P. Pfeiffer, Nikolas A. Wittek, Nils L. Vu, Kyle C. Nelli, Alexander Carpenter, Geoffrey Lovelace, Mark A. Scheel, and William Throwe. Scalarization of isolated black holes in scalar Gauss-Bonnet theory in the fixing-the-equations approach. *Phys. Rev. D*, 110(2):024033, 2024.

[14] Geoffrey Lovelace, Kyle C. Nelli, et al. Simulating binary black hole mergers using discontinuous Galerkin methods. *Class. Quant. Grav.*, 42(3):035001, 2025.

[15] Sizheng Ma et al. Fully relativistic three-dimensional Cauchy-characteristic matching for physical degrees of freedom. *Phys. Rev. D*, 109(12):124027, 2024.

[16] Sizheng Ma, Kyle C. Nelli, Jordan Moxon, Mark A. Scheel, Nils Deppe, Lawrence E. Kidder, William Throwe, and Nils L. Vu. Einstein–Klein–Gordon system via Cauchy-characteristic evolution: computation of memory and ringdown tail. *Class. Quant. Grav.*, 42(5):055006, 2025.

[17] Sizheng Ma, Mark A. Scheel, Jordan Moxon, Kyle C. Nelli, Nils Deppe, Lawrence E. Kidder, William Throwe, and Nils L. Vu. Merging black holes with Cauchy-characteristic matching: Computation of late-time tails. *Phys. Rev. D*, 112(2):024003, 2025.

[18] Lorena Magaña Zertuche et al. High-precision ringdown surrogate model for nonprecessing binary black holes. *Phys. Rev. D*, 112(2):024077, 2025.

[19] Keefe Mitman et al. Probing the ringdown perturbation in binary black hole coalescences with an improved quasinormal mode extraction algorithm. *Phys. Rev. D*, 112(6):064016, 2025.

[20] Kyle C. Nelli, William Throwe, Nils Deppe, Mark A. Scheel, Lawrence E. Kidder, Nils L. Vu, and Saul A. Teukolsky. Horizon tracking for asynchronous parallel black hole simulations. 8 2025.

[21] Milton Ruiz, Antonios Tsokaros, Stuart L. Shapiro, Kyle C. Nelli, and Sam Qunell. Magnetic Ergostars, Jet Formation and Gamma-Ray Bursts: Ergoregions versus Horizons. *Phys. Rev. D*, 102(10):104022, 2020.

[22] Mark A. Scheel et al. The SXS collaboration's third catalog of binary black hole simulations. *Class. Quant. Grav.*, 42(19):195017, 2025.

[23] Nikolas A. Wittek, Leor Barack, Harald P. Pfeiffer, Adam Pound, Nils Deppe, Lawrence E. Kidder, Alexandra Macedo, Kyle C. Nelli, William Throwe, and Nils L. Vu. Relieving Scale Disparity in Binary Black Hole Simulations. *Phys. Rev. Lett.*, 134(25):251402, 2025.

[24] Hengrui Zhu et al. Nonlinear effects in black hole ringdown from scattering experiments: Spin and initial data dependence of quadratic mode coupling. *Phys. Rev. D*, 109(10):104050, 2024.

[25] Hengrui Zhu et al. Black hole spectroscopy for precessing binary black hole coalescences. *Phys. Rev. D*, 111(6):064052, 2025.

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Chapter 1

INFRASTRUCTURE FOR CROSS-CODE COMPARISON OF CCE WAVEFORMS

1.1 Introduction

Being able to compare gravitational waveforms with each other, whether it be comparing Numerical Relativity (NR) waveforms with other NR waveforms or observed gravitational waves (GWs), is an extremely important ability in GW science; potentially the most important. GW detectors compare suspected signals to models that have been informed by NR such as the IMRPhenom family (examples: [1–3]) for frequency-domain analysis and the SEOBNR family (examples: [4–6]) for time-domain analysis. NR codes also compare waveforms internally, typically for verification of convergence as in §**??** or for analysis of trends in GW catalogs [7]. However, comparing waveforms from different NR simulations, even from the same code, have to be done carefully since different simulations will be done in different gauges.

1.1.1 Electricity and Magnetism example

To give an example of why we need to consider gauges when comparing NR waveforms, let us first look at a more familiar use of gauges in an introductory setting such as electricity and magnetism. Maxwell's equations (in vector form) are

$$\nabla \cdot \mathbf{E} = 4\pi\rho \qquad \nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t} \tag{1.1}$$

$$\nabla \cdot \mathbf{B} = 0 \qquad \nabla \times \mathbf{B} = \frac{1}{c}\left(4\pi\mathbf{J} + \frac{\partial \mathbf{E}}{\partial t}\right) \tag{1.2}$$

From Gauss's law for magnetism and Faraday's equation along with vector calculus identities, we can define

$$\mathbf{B} = \nabla \times \mathbf{A} \tag{1.3}$$

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t} \tag{1.4}$$

where ϕ is the electric potential and $\mathbf{A}$ is the vector potential. We notice that if we modify ϕ and $\mathbf{A}$ like so

$$\mathbf{A} \to \mathbf{A} + \nabla\psi \tag{1.5}$$

$$\phi \to \phi - \frac{\partial\psi}{\partial t} \tag{1.6}$$

for some scalar function ϕ, $\mathbf{B}$ and $\mathbf{E}$ in 1.3 and 1.4, respectively, remain unchanged. This ability to choose ϕ and $\mathbf{A}$ is called gauge freedom, and the choice that we make is considered the gauge.

In electricity and magnetism there are two common gauges; Coulomb and Lorenz. The Coulomb gauge is defined by

$$\nabla \cdot \mathbf{A} = 0 \tag{1.7}$$

To describe the Lorenz gauge, we combine ϕ and $\mathbf{A}$ as a four-vector

$$A^\mu = (\phi/c, \mathbf{A})\,. \tag{1.8}$$

The Lorenz gauge is then

$$\partial_\mu A^\mu = 0. \tag{1.9}$$

This differs from the Coulomb gauge because of the $\partial_0 A^0 \sim \partial_t \phi$ part. The implications that this choice of gauge has on the equations can be found in virtually any electricity and magnetism textbook [TODO:CITE]. The important point for this discussion is that even though $\mathbf{E}$ and $\mathbf{B}$ mathematically remain unchanged depending on which gauge is chosen, the gauge dependent quantities ϕ and $\mathbf{A}$ cannot be compared between gauges.

1.1.2 General and Numerical Relativity gauges

Now we move to gauges in general relativity (GR). Instead of putting a constraint on the four-potential as our gauge condition, in GR we impose our gauge condition on the coordinates x^μ of our spacetime. We skip a derivation of *why* we have this coordinate freedom in the theory of general relativity as that's the topic of introductory GR courses, but a derivation can be found in virtually any GR text book, including [8–10] to name a few. For our purposes, let's take a look at a few gauges in GR and NR.

1.1.2.1 Transverse-traceless gauge

The first gauge most students encounter is one called the transverse-traceless (TT) gauge, occasionally called the radiation gauge. The TT gauge is particularly useful

in a linearized theory of gravity where we define our metric as

$$g_{\mu\nu} = \eta_{\mu\nu} + \bar{h}_{\mu\nu} \tag{1.10}$$

with

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h \tag{1.11}$$

where $\eta_{\mu\nu}$ is our flat background spacetime metric and $h_{\mu\nu}$ is a small perturbation of this background. Here, $\bar{h}_{\mu\nu}$ is a "trace-reversed" form. We take our field equations

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi T_{\mu\nu} \tag{1.12}$$

and substitute Eq. 1.10 into them. This will contain second derivatives of $\bar{h}_{\mu\nu}$ because of the Ricci tensor and scalar. It is at this point where we make a gauge choice of

$$\Box x^{\mu} = 0 \tag{1.13}$$

which can be shown to simplify to [10]

$$\Box x^{\mu} = 0 \rightarrow g^{\mu\nu}\Gamma^{\rho}_{\mu\nu} = 0 \rightarrow \partial_{\mu}\bar{h}^{\mu}_{\lambda} = 0 \tag{1.14}$$

in the weak-field (linear) limit. Plugging Eq. 1.14 into Eq. 1.12 and taking the vacuum case ($R_{\mu\nu} = T_{\mu\nu} = 0$), we arrive at a clean result

$$\Box\bar{h}_{\mu\nu} = 0. \tag{1.15}$$

It can be shown (through some math that is unimportant to the point here) [10] that

$$h^{\mathrm{TT}}_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_{+} & h_{\times} & 0 \\ 0 & h_{\times} & -h_{+} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{1.16}$$

where we have dropped the bars because $h_{\mu\nu}$ is the trace-reverse of $\bar{h}_{\mu\nu}$ which is traceless. These are the familiar GW polarizations h_{+} and $h_{\times}$ for a linear GW propagating in the z-direction. Therefore, this is a useful gauge to analyze GWs from one of the detectors (though it's quite a bit more complicated than this in reality [11–14]).

1.1.2.2 NR gauges

Moving on to NR, the two most popular formulations of the field equations (Eq. 1.12) are the Generalized Harmonic (GH) formulation [15] and the BSSN/Z4 family [TODO:CITE]. Each of these formulations use different gauge conditions for the evolution of the coordinates x^μ, none that are equal to the TT gauge.

We start with GH. There are two dominant gauges used when evolving the GH formulation. The first is actually the harmonic gauge introduced in Eq. 1.13. The TT gauge is actually a sub-category of the harmonic gauge, useful for linear theories. However, GH is the fully non-linear theory of gravity and so the TT gauge is not suitable here. The harmonic gauge is convenient because the solutions for the equations of the coordinates are harmonic (or wave) solutions, hence the name, and we know how to solve for waves.

The second gauge commonly used in GH evolutions is one that actually builds off of the harmonic gauge, because it's wave solutions are so desirable. The general form of this gauge, called the damped-harmonic gauge, is

$$\Box x_\mu = H_\mu(x^\nu, g_{\nu\kappa}) \tag{1.17}$$

where H_μ is called the gauge-source function [16], and it is a function of the coordinates and the spacetime metric. The gauge source function is defined in Eq. **??–??**. The main purpose for this is to damp away oscillations shift α and drive it to one (so the time evolution remains steady).

For the BSSN/Z4 family of formulations, a gauge condition called the moving-puncture gauge, combined with $1 + \log$ slicing and a Γ-driver condition, is used. For this family, conditions are put onto the lapse function α and the shift vector β^i instead of having equations for the coordinates themselves. The $1 + \log$ slicing condition is (Ch. 4 of [17])

$$\partial_t \alpha = -2\alpha K + \beta^j \partial_j \alpha \tag{1.18}$$

where K is the extrinsic curvature. The Γ-driver condition can come in two forms (Ch. 4 of [17])

$$\partial_t \beta^i = \eta B^i + \beta^j \partial_j \beta^i, \tag{1.19}$$

or ([18])

$$\partial_t \beta^i = \tilde{\Gamma}^i - \eta \beta^i + \beta^j \partial_j \beta^i, \tag{1.20}$$

where η is a tunable parameter, and $\tilde{\Gamma}^i$ is related to the Christoffel symbols. The former is typically used with BSSN while the later is used with the Z4 family, but many NR codes make slight modifications to any of these gauge equations.

It should be fairly clear that trying to compare a GW produced with the GH formulation to a GW produced with the BSSN formulation is not well-defined because the gauges that are used are completely different. However, more subtly, comparing GWs that were produced with the same formulation is also ill-defined. This is due to the limitations of numerical resolution and floating-point arithmetic when solving Einstein's Equations on the computer. Because of these numerical issues, even simulations that start with the same initial data but are evolved with different spatial or temporal resolutions may not evolve in an identical manner. Therefore, the coordinates of these different resolution simulations will be different at the same "time" of the simulation. This can be seen in the bottom two panels of Figure **??** where we are comparing the phase and amplitude difference of the GWs from different resolutions of BBH simulations evolved with the same initial data.

1.2 CCE?

Now that we have detailed a number of different useful gauges, namely the TT, harmonic, damped harmonic, and moving-puncture gauges, it should be clear that the GW strains h we compute from simulations done in these different gauges cannot directly be compared with each other. The coordinate freedom of GR implies that even if we use the same formulas to compute h, we will get different answers because the coordinates are completely different. Therefore, how can we compare waveforms across codes? The answer is, by comparing the waveforms at a location where our choice of coordinates mostly don't matter; Namely at future-null infinity, also denoted $\mathscr{I}+$. Future-null infinity is the location in spacetime that future null rays (e.g. light rays) escape if they don't terminate inside the black hole and aren't in a stable orbit.

There are a few ways of computing GWs at $\mathscr{I}+$that allow for comparison of GWs between resolutions or between codes. Most NR codes don't have a computational domain that extends out to $\mathscr{I}+$, as this would be intractably computationally expensive with typical methods. Instead, GWs at $\mathscr{I}+$must be computed in another manner. The first, and by far most inaccurate, method is to simply ignore gauge effects and compare GWs that are extracted at finite radii within the computational domain, treating them as if they were at $\mathscr{I}+$. While this method cannot be used to

compare waveforms between NR codes, when comparing similar simulations from the same code, it can be a quick-and-dirty method for a quick analysis.

The second, and by far most common, method for computing GWs at $\mathscr{I}+$is called extrapolation. The details for extrapolation can be found in the SXS Collaboration's second catalog release [19]. However, the overall method is to compute the strain h and the Ψ_4 Weyl scalar on a number of concentric spheres surrounding the origin of the coordinate system. These quantities are then extrapolated with a $1/r^N$ power law, where N is chosen based on a number of different conditions described in [19].

Extrapolation has been very successful and has been widely used by the NR community for computing GWs [TODO:CITE]. However, because it simply extrapolates waveforms from close to the black hole, some non-linear effects of GR as the wave travels to $\mathscr{I}+$are not accounted for and may be important. Even with memory corrected waveforms from [7], they still don't account for all non-linear effects. For this, a full evolution of Einstein's equations out to $\mathscr{I}+$is needed.

1.2.1 Also CCE?

A method to evolve Einstein's equations all the way out to $\mathscr{I}+$is called Cauchy-Characteristic Extraction/Evolution (CCE). Here, we detail the important concepts and equations of CCE that will be needed in the rest of the chapter, following [20] and [21].

CCE works by first performing a Cauchy evolution of the spacetime using one of the formulations we discussed earlier this chapter (GH, BSSN, Z4). Then, we store the spacetime metric and its derivatives on a sphere centered at the origin for all times during our simulation. This is called the worldtube. This worldtube data is then used as the inner boundary condition for the Characteristic evolution which goes out to $\mathscr{I}+$. This characteristic evolution is performed on null-hypersurfaces, unlike the Cauchy evolution which is performed on space-like hypersurfaces. This is because with the Characteristic evolution, we are primarily concerned with evolving and resolving GWs, which travel along null rays, making null-hypersurfaces well suited to this problem.

CCE decomposes the metric into again different coordinates and gauge conditions that GH or BSSN. The metric in a Bondi-like spherical coordinate system $\{u, r, x^A\}$

is

$$ds^2 = -\left(e^{2\beta}\frac{V}{r} - r^2 h_{AB}U^A U^B\right)du^2 - 2e^{2\beta}dudr - 2r^2 h_{AB}U^B dudx^A + r^2 h_{AB}dx^A dx^B \tag{1.21}$$

where h_{AB} is the related to the metric on the 2-sphere q_{AB}, V and β are scalar functions, and U^A is a vector on the 2-sphere. These Bondi-like coordinates have the gauge conditions that $g_{rr} = g_{rA} = 0$, $\det(h_{AB}) = \det(q_{AB})$, and asymptotic behavior

$$\lim_{x\to\infty}\{\beta, U^A, h_{AB}, \frac{V-r}{r^2}\} = \mathcal{O}(r^0). \tag{1.22}$$

We can define a complex dyad q^A (basis vector) of q_{AB} as the following

$$q_{AB} = \frac{1}{2}\left(q_A\bar{q}_B + \bar{q}_A q_B\right), \tag{1.23}$$

with barred quantities being the complex conjugate. Additionally, if a quantity transforms like

$$q_A \to q_A e^{is\psi} \tag{1.24}$$

for some phase ψ, then we say that quantity has spin weight s. With these definitions, it is now convenient to define some scalars that will be important for the CCE evolution:

$$U \equiv U^A q_A \qquad (s=1) \tag{1.25}$$

$$Q \equiv r^2 e^{-2\beta} q^A h_{AB}\partial_r U^B \qquad (s=1) \tag{1.26}$$

$$r^2 W \equiv V - r \qquad (s=0) \tag{1.27}$$

$$J \equiv \frac{1}{2}q^a q^B h_{AB} \qquad (s=2) \tag{1.28}$$

$$K \equiv \sqrt{1 - J\bar{J}} \qquad (s=0) \tag{1.29}$$

$$H \equiv \partial_u J \qquad (s=2) \tag{1.30}$$

Now we can define a hierarchical set of evolution equations for these scalars which will give us the characteristic evolution of the spacetime on null-slices. The first of these equations that is needed is actually Eq. 1.30, which will give us J. Then once we have J, we can build up a solution by solving radial differential equations for the other scalars. Here we will denote the right hand side of each equation with a generic function S because the details of what is on the right hand side are not

important. See [20] again for the full equations.

$$\partial_r \beta = S_\beta(J), \tag{1.31}$$

$$\partial_r (r^2 Q) = S_Q(J, \beta), \tag{1.32}$$

$$\partial_r U = S_U(J, \beta, Q), \tag{1.33}$$

$$\partial_r (r^2 W) = S_W(J, \beta, Q, U), \tag{1.34}$$

$$\partial_r (rH) + L_H(J, \beta, Q, U, W) H \tag{1.35}$$

$$+ L_{\bar{H}}(J, \beta, Q, U, W) \bar{H} = S_H(J, \beta, Q, U, W) \tag{1.36}$$

The final radial equation for H lets us solve Eq. 1.30 for J at the next time step and the hierarchy starts over again.

Now these scalars are defined in terms of our metric expressed in a Bondi-like gauge which is useful for evolving the Characteristic system. However, in order for us to impose our worldtube data which is in some arbitrary evolved gauge, we must transform the evolved gauge into our Bondi-like gauge. The mathematics of this transformation are quite tedious and unimportant for the point being made here because it uses a number of intermediate coordinates that are only helpful for the transformation, so we will skip them. To see them in detail, refer to [20]. Instead we just state that our scalars defined in 1.25-1.30 can be expressed as some function of the Cauchy spacetime metric and Cauchy evolved coordinates (e.g. $U = f(x^\mu, g_{\mu\nu})$, where x^μ is not related to x^A), with all the coordinate transformations hidden.

1.2.2 Machinery, Figure out section name later

Now we have the basics of the Characteristic evolution. Specifically that we use the Bondi scalars defined in Eqs. 1.25-1.30 as part of a hierarchical evolution system, we need those Bondi scalars on the worldtube which serve as the inner boundary condition for the Characteristic evolution, and these Bondi scalars on the worldtube can be computed from the Cauchy metric, coordinates, and the spatial and temporal derivatives of the metric. Next we focus on how this last step is implemented in SpECTRE and what new features were added in SpECTRE.

The first detail we need to discuss is how SpECTRE actually represents and stores the Bondi scalars either in memory of the simulation or on disk. Since we need the Bondi scalars on a sphere of constant radius at each timestep, representing the scalars as complex spherical harmonic coefficients is the most natural. For a surface $S_{\ell,m}$, we require $m = \ell$. This implies we will use $M = (\ell + 1)^2$ complex-valued coefficients (or modes) to represent each Bondi scalar. We will call this the "modal"

representation. The modal representation is extremely useful for two reasons. First is that it is straight forward to interpolate between two representations with different ℓ since the interpolation is either truncating the number of coefficients or appending coefficients with value zero. Second, is that the modal representation is significantly more compact that a "nodal" representation where we represent the complex valued scalar at collocation points on a sphere. For the same surface $S_{\ell,m}$ the number of nodes needed to represent the scalar is $N = (\ell + 1)(2\ell + 1) > M$. The advantage with using a nodal representation for the actual Characteristic evolution is that we can easily perform arithmetic between Bondi scalars which is needed for the spectral algorithms that SpECTRE uses. This is significantly more difficult with a modal representation because you cannot perform arithmetic with coefficients.

We now have a situation where it is ideal to represent the evolved Bondi scalars with nodal values, but the worldtube Bondi variables with modal coefficients. The worldtube variables must be written to disk during the Cauchy evolution, so we want those to take up as little disk space as possible. Also, there's no guarantee that the ℓ we write the worldtube variables with will be the same ℓ that we use for the inner boundary of our Characteristic evolution. One may even want to run multiple Characteristic evolutions with varying resolution.

The SpECTRE CCE algorithm [21] was previously designed to accept two different forms of the Bondi variables on the worldtube. The first is reading in α, β^i, g_{ij}, and their derivatives each written out in complex modal form. This was to support a legacy output format from the SpEC code [22]. The second is directly reading in the Bondi scalars in complex modal form in the Cauchy evolution gauge. This format is output from the modern SpEC simulations [7], and from all SpECTRE simulations [23]. Both SpEC and SpECTRE have routines within the Cauchy evolution that compute the necessary Bondi scalars in the evolution gauge on the worldtube. The SpECTRE CCE algorithm then reads in both these data formats. For the metric variables, a conversion to Bondi scalars is first computed within the CCE algorithm.

1.2.3 New input formats to SpECTRE CCE

In this work, we have added three more input formats for SpECTRE CCE. They are 1) Bondi scalars in complex nodal format, 2) metric variables in complex modal format but without needing the time derivatives of the metric quantities, and 3) metric variables in complex nodal format, both with and without the time derivatives of the metric quantities. We have added support for reading in nodal data formats

because it is often easier for other NR codes besides SpEC and SpECTRE to output data onto a set of grid points rather than transforming the data to the exact form of complex spherical harmonic coefficients that SpECTRE requires. Our spherical harmonic formulation, however, does require the worldtube data to be output at specific grid, or collocation, points necessary for the CCE algorithm. To aid in giving the correct collocation points, SpECTRE offers a separate executable that will write the correct collocation points for a given radius to a text file in the necessary order. SpECTRE also offers a C routine that can be added to the source code that will give an array of the necessary collocation points.

We have expanded support for reading in metric quantities for the worldtube because most other NR codes use either the BSSN or Z4 formulations of Einstein's equations. Particularly we have relaxed the requirement to provide time derivatives of the metric quantities for the worldtube data. This was done because unlike the GH formulation where one of the evolved variables, Π, *is* the time derivative of the spacetime metric, the BSSN and Z4 formulations only evolve the metric itself. Even though the time derivative of the metric quantities are computed for the Cauchy evolution, they are often not available for output. The time derivatives within the evolution would be out-of-sync with the actual evolved metric quantities by one time step and trying to sync up the data is a much larger change for other codes.

Therefore, we allow for computing the time derivatives internally in SpECTRE CCE. When doing this, extreme care must be taken so that we use the correct gauge equations for α and β^i. Luckily for α, most other codes use the $1 + \log$ slicing condition given by Eq. 1.18. Some codes choose to not include the advective term $\beta^j \partial_j \alpha$, so we have added it as optional. For the gamma-driver condition, we support both Eqs. 1.19 and 1.20. Again, we make the advective term $\beta^j \partial_j \beta^i$ optional. For the different gamma-driver conditions, they require different variables to be written to disk from the Cauchy evolution. This is the responsibility of the user to know which variables need to be written out.

Due to the increased number of supported worldtube data formats, the inputs to the actual SpECTRE CCE algorithm have been simplified. We now restrict the worldtube input data format to only be the Bondi variables represented in complex modal coefficients. This was done largely to avoid waiting for extra computations during the Characteristic evolution for the worldtube data. It was also done to simplify the code-paths taken during the Characteristic evolution. Therefore, in order to support all these worldtube data formats, SpECTREnow offers a separate

executable that will transform the worldtube data from any of the supported input formats to the Bondi scalars in complex modal form in a separate output file. The new output file is then fed directly into the CCE algorithm. The new executable has the ability to handle overlapping output times and multiple output files from different checkpointed segments of a simulation.

Finally, in order to make these new improvements available to the NR community, `SpECTRE` now offers a downloadable `tar` file which has the following pre-built executables:

1. The CCE executable itself plus an associated input file.
2. A preprocessing executable to convert the worldtube data into the Bondi modal format plus an associated input files with examples for each of the accepted input data types.
3. An executable to write the collocation points for a sphere at a given radius and angular resolution ℓ to interpolate input data to.
4. Testing scripts plus a small amount of example worldtube data used to verify the executables work.

Additionally, we have added detailed documentation to the `SpECTRE` webpage with instructions for how to use these executables, recommended settings for certain input parameters, and other small pieces of knowledge that were empirically determined while running CCE.

1.2.4 Frame fixing and BMS transformations

1. CCE isn't the final step
2. At $\mathscr{I}+$, we have group freedom (BMS)
3. Explain supertranslations (and rotations?)
4. Briefly explain memory
5. Show examples of (2,0) mode memory correction

1.2.5 Examples of NR codes using SpECTRE CCE

1. SpECTRE,SpEC
2. Hengrui, Athenak
3. Deborah, Maya/ETK
4. Dongze/Elias, FIL
5. Sarah's paper??
6. Hengrui did study with same ID run through AthenaK and SpECTRE?
7. If not, that could be "future work"
8. Future work analyzing the differences in behavior of waveforms produced by different codes.

1. Motivation
   a) Explain importance about comparing waveforms
   b) Waveform models, surrogates, etc.
   c) h_+ and $h_\times$ not gauge invariant
   d) Analog to 4 potential from E&M
   e) Different evolution gauges
2. Overview of CCE (from Jordan's paper)
3. Different formulations evolved variable
4. Converting to "standard" variables for CCE
5. Caveat of frame fixing/BMS transforms
6. Examples of other codes using this

References

Bibliography

[1] Geraint Pratten, Sascha Husa, Cecilio Garcia-Quiros, Marta Colleoni, Antoni Ramos-Buades, Hector Estelles, and Rafel Jaume. Setting the cornerstone for a family of models for gravitational waves from compact binaries: The dominant harmonic for nonprecessing quasicircular black holes. *Phys. Rev. D*, 102(6):064001, 2020. doi: 10.1103/PhysRevD.102.064001.

[2] Cecilio García-Quirós, Marta Colleoni, Sascha Husa, Héctor Estellés, Geraint Pratten, Antoni Ramos-Buades, Maite Mateu-Lucena, and Rafel Jaume. Multimode frequency-domain model for the gravitational wave signal from nonprecessing black-hole binaries. *Phys. Rev. D*, 102(6):064002, 2020. doi: 10.1103/PhysRevD.102.064002.

[3] Geraint Pratten et al. Computationally efficient models for the dominant and subdominant harmonic modes of precessing binary black holes. *Phys. Rev. D*, 103(10):104056, 2021. doi: 10.1103/PhysRevD.103.104056.

[4] Mohammed Khalil, Alessandra Buonanno, Hector Estelles, Deyan P. Mihaylov, Serguei Ossokine, Lorenzo Pompili, and Antoni Ramos-Buades. Theoretical groundwork supporting the precessing-spin two-body dynamics of the effective-one-body waveform models SEOBNRv5. *Phys. Rev. D*, 108(12): 124036, 2023. doi: 10.1103/PhysRevD.108.124036.

[5] Antoni Ramos-Buades, Alessandra Buonanno, Héctor Estellés, Mohammed Khalil, Deyan P. Mihaylov, Serguei Ossokine, Lorenzo Pompili, and Mahlet Shiferaw. Next generation of accurate and efficient multipolar precessing-spin effective-one-body waveforms for binary black holes. *Phys. Rev. D*, 108(12): 124037, 2023. doi: 10.1103/PhysRevD.108.124037.

[6] Maarten van de Meent, Alessandra Buonanno, Deyan P. Mihaylov, Serguei Ossokine, Lorenzo Pompili, Niels Warburton, Adam Pound, Barry Wardell, Leanne Durkan, and Jeremy Miller. Enhancing the SEOBNRv5 effective-one-body waveform model with second-order gravitational self-force fluxes. *Phys. Rev. D*, 108(12):124038, 2023. doi: 10.1103/PhysRevD.108.124038.

[7] Mark A. Scheel et al. The SXS collaboration's third catalog of binary black hole simulations. *Class. Quant. Grav.*, 42(19):195017, 2025. doi: 10.1088/1361-6382/adfd34.

[8] Charles W. Misner, K. S. Thorne, and J. A. Wheeler. *Gravitation*. W. H. Freeman, San Francisco, 1973. ISBN 978-0-7167-0344-0, 978-0-691-17779-3.

[9] Robert M. Wald. *General Relativity*. Chicago Univ. Pr., Chicago, USA, 1984. doi: 10.7208/chicago/9780226870373.001.0001.

[10] Sean M. Carroll. *Spacetime and Geometry: An Introduction to General Relativity*. Cambridge University Press, 7 2019. ISBN 978-0-8053-8732-2, 978-1-108-48839-6, 978-1-108-77555-7. doi: 10.1017/9781108770385.

[11] R. Abbott et al. GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo during the Second Part of the Third Observing Run. *Phys. Rev. X*, 13(4):041039, 2023. doi: 10.1103/PhysRevX.13.041039.

[12] R. Abbott et al. GWTC-2.1: Deep extended catalog of compact binary coalescences observed by LIGO and Virgo during the first half of the third observing run. *Phys. Rev. D*, 109(2):022001, 2024. doi: 10.1103/PhysRevD.109.022001.

[13] R. Abbott et al. GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run. *Phys. Rev. X*, 11: 021053, 2021. doi: 10.1103/PhysRevX.11.021053.

[14] B. P. Abbott et al. GWTC-1: A Gravitational-Wave Transient Catalog of Compact Binary Mergers Observed by LIGO and Virgo during the First and Second Observing Runs. *Phys. Rev. X*, 9(3):031040, 2019. doi: 10.1103/PhysRevX.9.031040.

[15] Lee Lindblom, Mark A. Scheel, Lawrence E. Kidder, Robert Owen, and Oliver Rinne. A New generalized harmonic evolution system. *Class. Quant. Grav.*, 23:S447–S462, 2006. doi: 10.1088/0264-9381/23/16/S09.

[16] Bela Szilagyi, Lee Lindblom, and Mark A. Scheel. Simulations of Binary Black Hole Mergers Using Spectral Methods. *Phys. Rev. D*, 80:124010, 2009. doi: 10.1103/PhysRevD.80.124010.

[17] Thomas W. Baumgarte and Stuart L. Shapiro. *Numerical Relativity: Solving Einstein's Equations on the Computer*. Cambridge University Press, 2010. doi: 10.1017/CBO9781139193344.

[18] David Hilditch, Sebastiano Bernuzzi, Marcus Thierfelder, Zhoujian Cao, Wolfgang Tichy, and Bernd Bruegmann. Compact binary evolutions with the Z4c formulation. *Phys. Rev. D*, 88:084057, 2013. doi: 10.1103/PhysRevD.88.084057.

[19] Michael Boyle et al. The SXS Collaboration catalog of binary black hole simulations. *Class. Quant. Grav.*, 36(19):195006, 2019. doi: 10.1088/1361-6382/ab34e2.

[20] Jordan Moxon, Mark A. Scheel, and Saul A. Teukolsky. Improved Cauchy-characteristic evolution system for high-precision numerical relativity waveforms. *Phys. Rev. D*, 102(4):044052, 2020. doi: 10.1103/PhysRevD.102.044052.

[21] Jordan Moxon, Mark A. Scheel, Saul A. Teukolsky, Nils Deppe, Nils Fischer, Francois Hébert, Lawrence E. Kidder, and William Throwe. SpECTRE Cauchy-characteristic evolution system for rapid, precise waveform extraction. *Phys. Rev. D*, 107(6):064013, 2023. doi: 10.1103/PhysRevD.107.064013.

[22] Nicholas W. Taylor, Michael Boyle, Christian Reisswig, Mark A. Scheel, Tony Chu, Lawrence E. Kidder, and Béla Szilágyi. Comparing Gravitational Waveform Extrapolation to Cauchy-Characteristic Extraction in Binary Black Hole Simulations. *Phys. Rev. D*, 88(12):124010, 2013. doi: 10.1103/PhysRevD.88.124010.

[23] Nils Deppe, William Throwe, Lawrence E. Kidder, Nils L. Vu, Kyle C. Nelli, Cristóbal Armaza, Marceline S. Bonilla, François Hébert, Yoonsoo Kim, Prayush Kumar, Geoffrey Lovelace, Alexandra Macedo, Jordan Moxon, Eamonn O'Shea, Harald P. Pfeiffer, Mark A. Scheel, Saul A. Teukolsky, Nikolas A. Wittek, Isha Anantpurkar, Carter Anderson, Michael Boyle, Alexander Carpenter, Andrea Ceja, Himanshu Chaudhary, Nicholas Corso, Nora Fayyazuddin Ljungberg, Francois Foucart, Noora Ghadiri, Matthew Giesler, Jason S. Guo, Sarah Habib, Chenhang Huang, Dante A. B. Iozzo, Ken Z. Jones, Guillermo Lara, Isaac Legred, Dongjun Li, Sizheng Ma, Denyz Melchor, Iago Mendes, Marlo Morales, Elias R. Most, Michael Murphy, Peter James Nee, Alejandro Osorio, Michael A. Pajkos, Kyle Pannone, Vaishak Prasad, Teresita Ramirez, Noah Ring, Hannes R. Rüter, Jennifer Sanchez, Leo C. Stein, Daniel Tellez, Sierra Thomas, Vittoria Tommasini, Daniel Vieira, Tom Wlodarczyk, David Wu, and Jooheon Yoo. Spectre, August 2025. URL `https://doi.org/10.5281/zenodo.16906840`.